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Pulmonary Infectious Diseases

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Reinvent the Future.

Session Objectives

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1. This presentation will offer you the opportunity to become familiar with the mechanisms leading to (community acquired) bacterial pneumonia and (community acquired) interstitial pneumonia and their morphology.

2. You will be able to define health care (nosocomial pneumonia), aspiration pneumonia including lung abscess.

3. In addition, you will be able to discuss and explain the morphology of chronic pneumonia caused by Mycobacterium tuberculosis, Histoplasma capsulatum, Blastomyces dermatitides and Coccidioides immitis.

3. Also, you will be able to describe morphology of the most common forms of pneumonia caused by opportunistic agents in HIV patients.

Please note that unless mentioned otherwise, all the images and information included in this presentation are from Robbins and Cotran, Pathologic Basis of Disease, 10th edition

Pneumonia

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Pneumonia is an infection of the lung parenchyma :

a. Lowered (decreased) systemic resistance of the host :

- chronic diseases, immunologic deficiencies, leukopenia
- treatment with immunosuppressive agents

b. Impaired local defense mechanisms:

• **Loss or suppression of the cough reflex**: altered sensorium (e.g., coma), anesthesia, neuromuscular disorders, leading to aspiration of gastric contents.

• **Dysfunction of the mucociliary apparatus**: cigarette smoke, inhalation of hot or corrosive gases, viral diseases, or genetic defects of ciliary function (e.g., immotile cilia syndrome).

• **Accumulation of secretions** : cystic fibrosis and bronchial obstruction.

• **Interference with the phagocytic and bactericidal activities of alveolar macrophages** by alcohol, tobacco smoke, anoxia, or oxygen intoxication.

• **Pulmonary congestion and edema**.

Classification

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Community-Acquired Acute Pneumonia
Streptococcus pneumoniae
Haemophilus influenzae
Moraxella catarrhalis
Staphylococcus aureus
Legionella pneumophila
Enterobacteriaceae *(Klebsiella pneumoniae)* and *Pseudomonas* spp.
Mycoplasma pneumoniae
Chlamydia spp. *(C. pneumoniae, C. psittaci, C. trachomatis)*
Coxiella burnetii (Q fever)
Viruses: respiratory syncytial virus, parainfluenza virus, and human metapneumovirus (children); influenza A and B (adults); adenovirus (military recruits)
Health Care–Associated Pneumonia
Staphylococcus aureus, methicillin-sensitive
Staphylococcus aureus, methicillin-resistant
Pseudomonas aeruginosa
Streptococcus pneumoniae
Hospital-Acquired Pneumonia
Gram-negative rods, Enterobacteriaceae (*Klebsiella* spp., *Serratia marcescens, Escherichia coli*) and *Pseudomonas* spp.
Staphylococcus aureus (usually methicillin-resistant)
Aspiration Pneumonia
Anaerobic oral flora *(Bacteroides, Prevotella, Fusobacterium, Peptostreptococcus)*, admixed with aerobic bacteria *(Streptococcus pneumoniae, Staphylococcus aureus, Haemophilus influenzae, Pseudomonas aeruginosa)*

Chronic Pneumonia
Nocardia
Actinomyces
Granulomatous: *Mycobacterium tuberculosis* and atypical mycobacteria, *Histoplasma capsulatum, Coccidioides immitis, Blastomyces dermatitidis*
Necrotizing Pneumonia and Lung Abscess
Anaerobic bacteria (extremely common), with or without mixed aerobic infection
Staphylococcus aureus, Klebsiella pneumoniae, Streptococcus pyogenes, and type 3 pneumococcus (uncommon)
Pneumonia in the Immunocompromised Host
Cytomegalovirus
Pneumocystis jiroveci
Mycobacterium avium-intracellulare complex
Invasive aspergillosis
Invasive candidiasis

Bacterial Pneumonia

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The alveoli show an inflammatory exudate, causing consolidation ("solidification") of the pulmonary tissue.

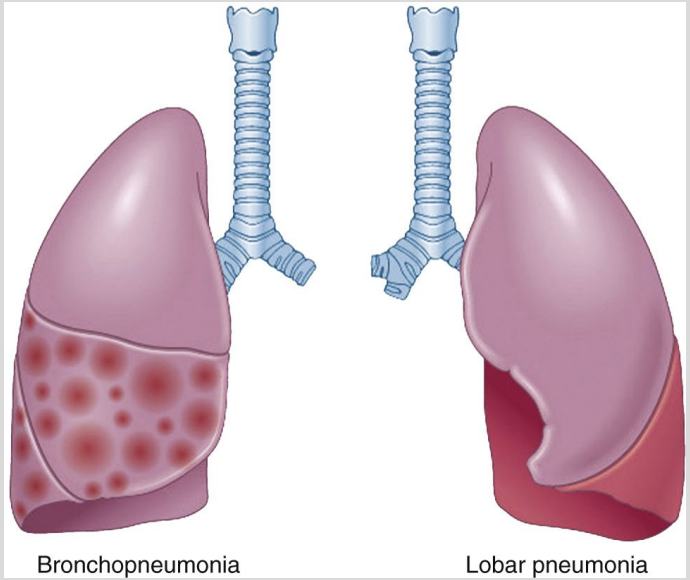
Predisposing conditions:
-extremes of age, chronic diseases (congestive heart failure, COPD, and diabetes),
-congenital or acquired immune deficiencies
-decreased or absent splenic function: risk for infection with encapsulated bacteria such as pneumococcus.

Clinical Features
-abrupt onset of high fever, shaking chills, and cough producing mucopurulent sputum and occasionally hemoptysis.
-If there is associated **pleuritis** the patient will develop pleuritic pain (on inspiration) and pleural friction rub.

The clinical picture is markedly modified by the administration of effective antibiotics. Appropriately treated patients may become afebrile with few clinical signs 48 to 72 hours after the initiation of antibiotics. The identification of the organism and the determination of its antibiotic sensitivity are the keystones of therapy

Bacterial Pneumonia Morphology

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Bronchopneumonia:
patchy consolidation; extension of previous bronchitis or bronchiolitis; seen in infancy and old age
The patchy involvement may become confluent, producing lobar consolidation.
Lobar Pneumonia: consolidation of a large portion of a lobe or entire lobe or the whole lung.

The same organisms may produce either pattern depending on patient susceptibility.

Lobar pneumonia	*S pneumoniae* (most common), *Legionella*, *Klebsiella*	Intra-alveolar exudate → consolidation A; may involve entire lobe B or the whole lung.
Bronchopneumonia	*S pneumoniae*, *S aureus*, *H influenzae*, *Klebsiella*	Acute inflammatory infiltrates C from bronchioles into adjacent alveoli; patchy distribution involving ≥ 1 lobe D.

First Aid Step I USMLE, 2022

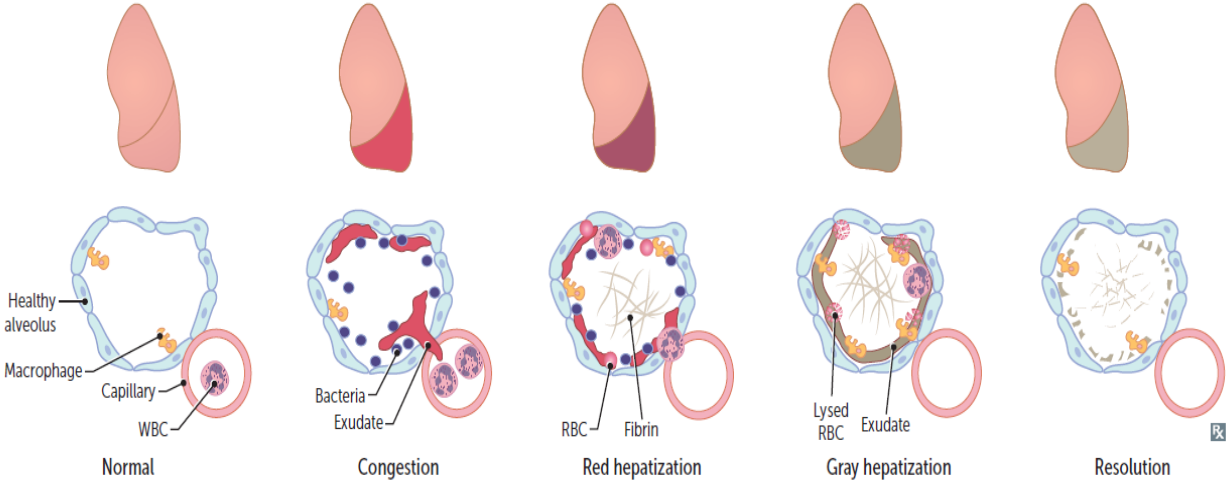
Natural history of lobar pneumonia

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Natural history of lobar pneumonia

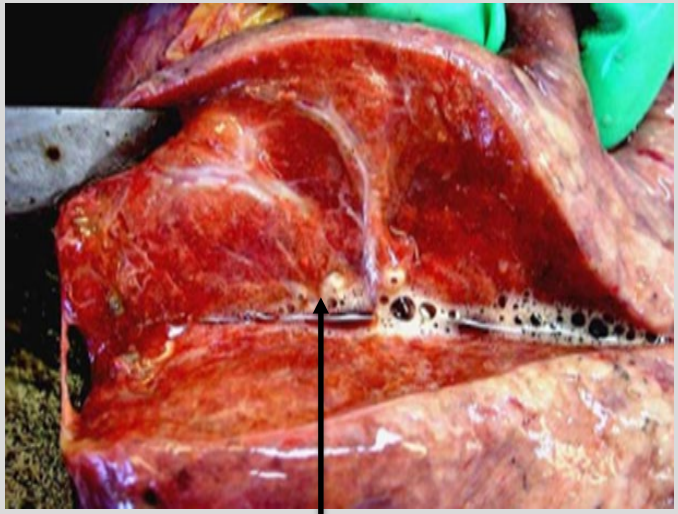
	Congestion	Red hepatization	Gray hepatization	Resolution
DAYS	1–2	3–4	5–7	8+
FINDINGS	Red-purple, partial consolidation of parenchyma Exudate with mostly bacteria	Red-brown consolidation Exudate with fibrin, bacteria, RBCs, WBCs Reversible	Uniformly gray Exudate full of WBCs, lysed RBCs, and fibrin	Enzymatic digestion of exudate by macrophages



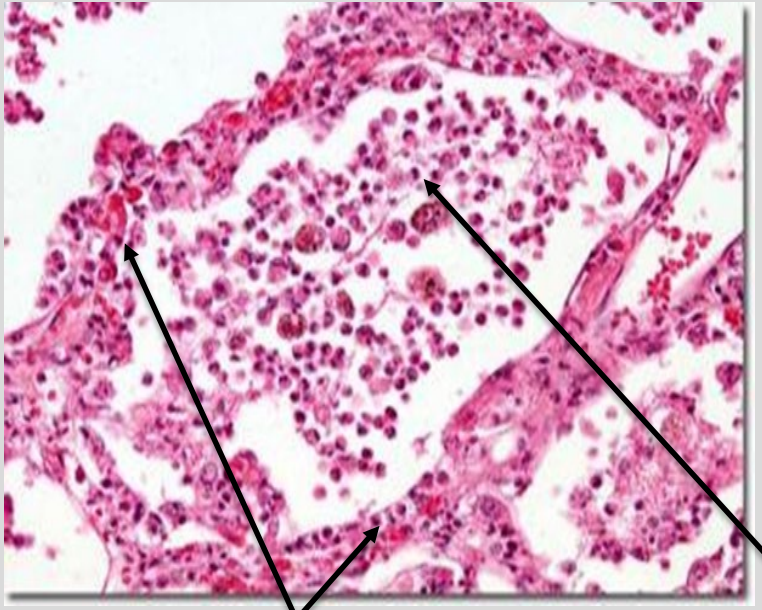
Red Hepatization

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Congestion of lung

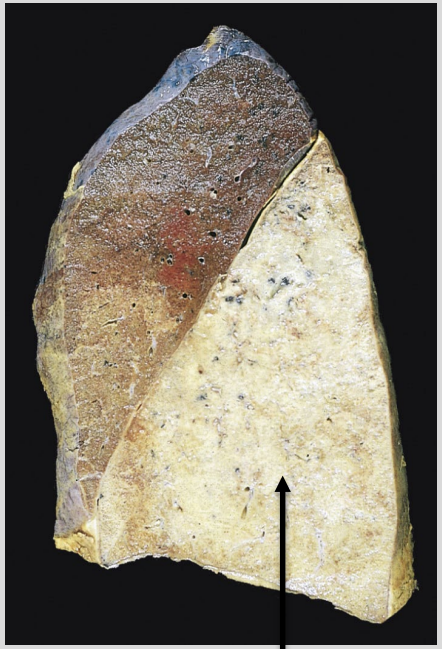


The congested septal capillaries and numerous intra-alveolar neutrophils.
Fibrin nets have not yet formed.

Grey Hepatization

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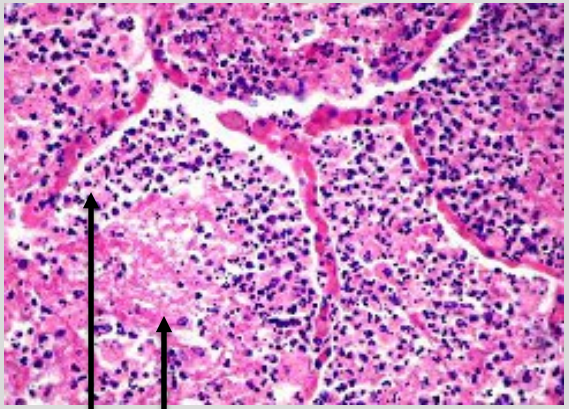
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The lower lobe is uniformly consolidated



Lobar Pneumonia: The middle and lower lobes of the right lung on the cut surface appear solid, airless, friable with a uniform pale white to focal pale red appearance (Stage of Grey Hepatization)



Exudate full of neutrophils and fibrin strands deposition

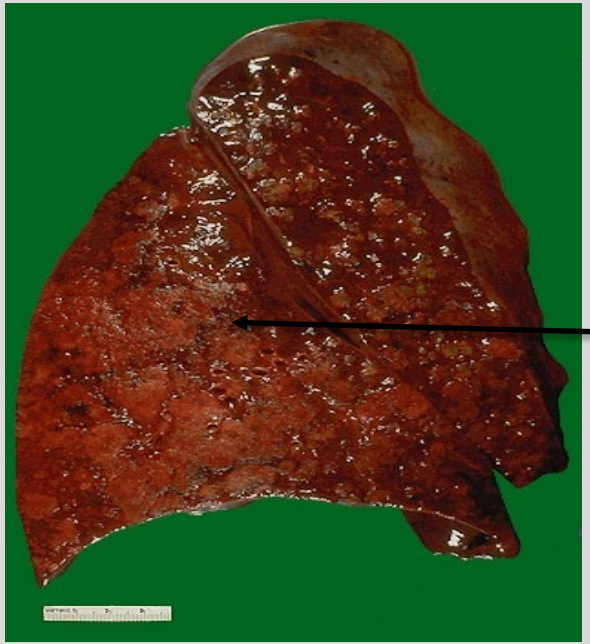
Bronchopneumonia Morphology

It is characterized by patchy areas of pulmonary consolidation, which are firmer than the surrounding lung. The inflammatory exudate may be confined to one lobe or may be multi-lobar.

Well-developed lesions: slightly elevated, dry, granular, gray-red to yellow, and poorly delimited at their margins

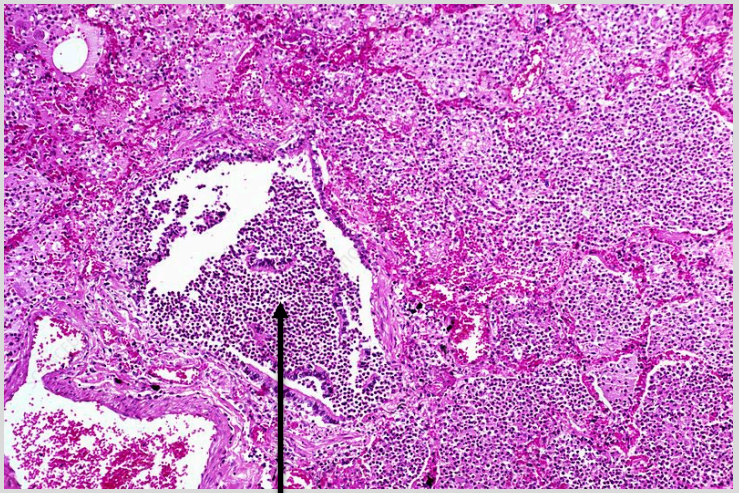
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Webpath

Cut surface of lung shows areas of tan-yellow consolidation.



Histologically, the reaction usually elicits a neutrophil-rich exudate that fills the bronchi, bronchioles, and adjacent alveolar spaces

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Bronchopneumonia Complications:

-Tissue destruction and necrosis causing abscess formation
-Empyema which represents spread of inflammation and destruction to pleura
-Organization of exudate to solid tissue
-Bacteremia with dissemination to heart valves, pericardium, brain, kidneys, spleen, joints

Interstitial (atypical) Pneumonia

Interstitial (atypical) pneumonia	*Mycoplasma*, *Chlamydophila pneumoniae*, *Chlamydophila psittaci*, *Legionella*, *Coxiella burnetii*, viruses (RSV, CMV, influenza, adenovirus)	Diffuse patchy inflammation localized to interstitial areas at alveolar walls; CXR shows bilateral multifocal opacities E. Generally follows a more indolent course ("walking" pneumonia).

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Atypical = moderate amount of sputum, no physical findings of consolidation, lack of alveolar exudate

Viral Pneumonia

Viruses have epithelial respiratory cells tropism.

Viral replication and gene expression leads to cytopathic changes, inducing **cell death and secondary inflammation**. The resulting damage and impairment of **local pulmonary defenses**, such as mucociliary clearance, may predispose to bacterial superinfections, which are often more serious than the viral infection itself.

Factors that favor extension of the infection to the lung :

- Extremes of age, Malnutrition, Alcoholism

Viral Pneumonia: Morphology

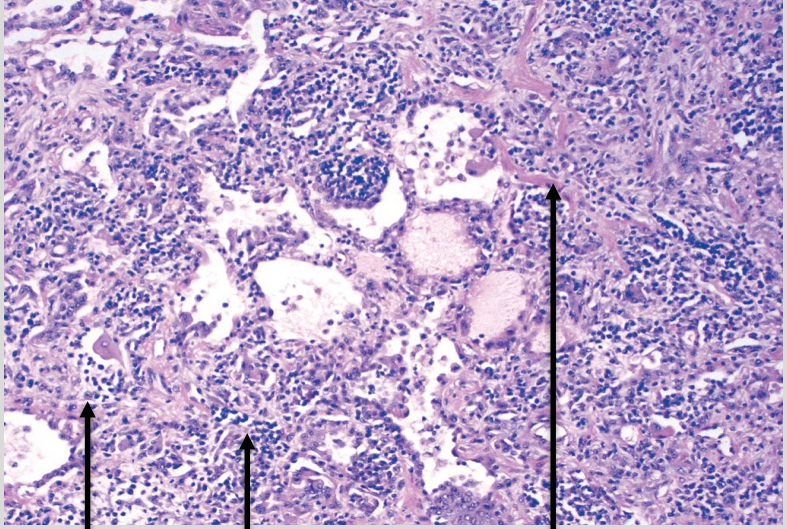
All viral infections produce similar morphologic changes

Upper respiratory infections :
- mucosal hyperemia and swelling
-overproduction of mucus secretions

The swollen mucosa and viscous exudate may plug the nasal channels, sinuses, or the Eustachian tubes, leading to **secondary suppurative bacterial infection**.

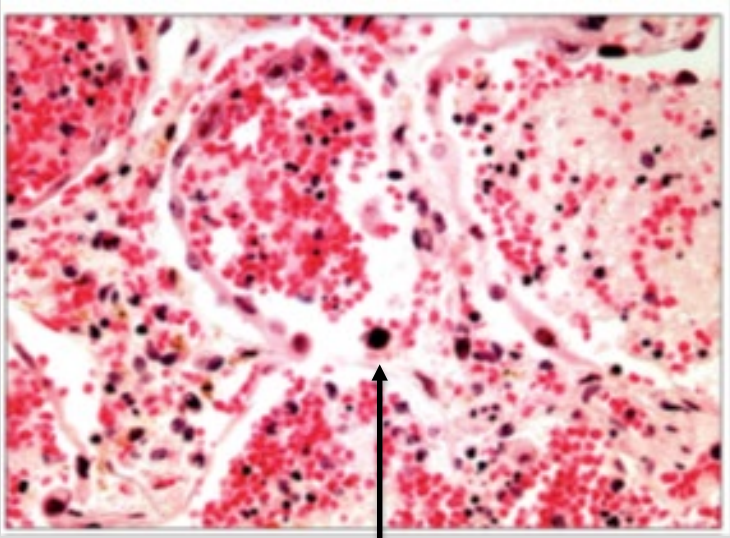
Children: **virus-induced tonsillitis** causing hyperplasia of the lymphoid tissue within the Waldeyer ring

Macroscopic lung involvement :
-patchy or may involve whole lobes bilaterally or unilaterally
-the affected areas are red-blue and congested
Pleuritis or pleural effusions are infrequent



Histology
-Interstitial inflammatory reaction in the walls of the alveoli.
-The alveolar septa are thick: mononuclear inflammatory infiltrate.

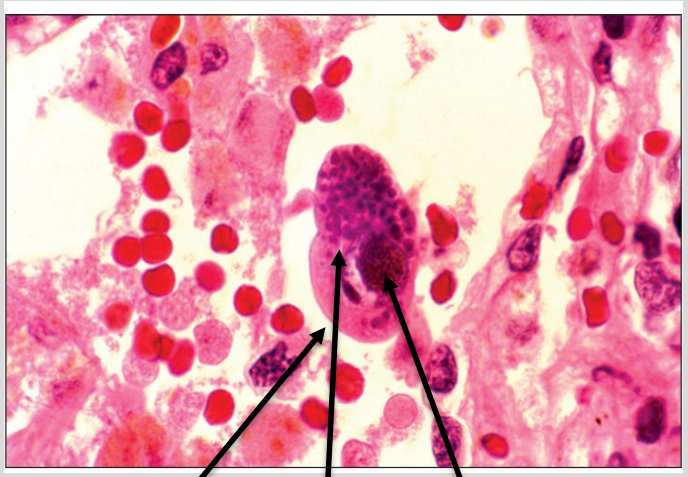
- If associated with ARDS, pink hyaline membranes line the alveolar walls.



Adenovirus

Necrotizing pneumonia with cellular debris in alveolar spaces

- Viral inclusions: dark nucleus in the center of this image



Cytomegalovirus

Cytomegaly with basophilic cytoplasmic inclusions and nuclear inclusions

Viral Pneumonia: Influenza type A

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Represents a major cause of pandemic and epidemic influenza.
: humans, pigs, horses, and birds

Influenza A has a viral envelope containing 2 proteins: hemagglutinin and neuraminidase.

Hemagglutinin: viral attachment
Neuraminidase: promote release of newly formed virions

Cellular destruction release chemokines and cytokines which will exacerbate inflammation.

Influenza type A:

- **antigenic drift:** mutations of the hemagglutinin and neuraminidase which allow virus to escape host antibodies.
- **antigenic shift:** hemagglutinin and neuraminidase are replaced through recombination of RNA segments with other viruses or dif strains of influenza A

Influenza type B and C: No antigenic drift and shift, mostly peds

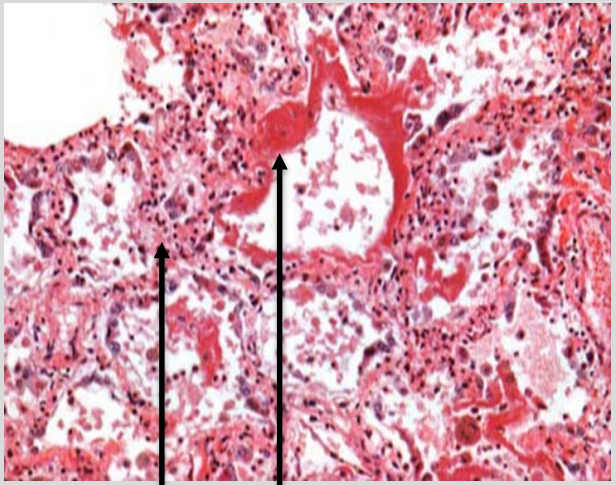
Influenza type A: Morphology

Mucosa: hyperaemia and swelling
Submucosa: lymphocytes, monocytes, and plasma
Overproduction of mucus, bacterial superinfection and ARDS may occur

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Macroscopic: lungs are congested, heavier and redder than normal and uniformly firm throughout.



Microscopic: Alveolar walls are thickened, and there is deposition of hyaline membranes

Interstitial (atypical) Pneumonia: Mycoplasma pneumoniae

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It is common among children and young adults
It may occur sporadically or as local epidemics in closed communities

Mp has adherence proteins;
It secretes hydrogen peroxide damaging the respiratory epithelium

Activation of cytokines and other inflammatory factors

IgM and IgG antibodies act as autoantibodies that cross react with neural cells and RBC

Clinical
Fever without chills
Dry cough: inhibition of ciliary movement
Diffuse crackles

Complications

Hemolysis

ITP
Acute hepatitis
Arthritis
Transverse myelitis
GI symptoms
Heart Disease

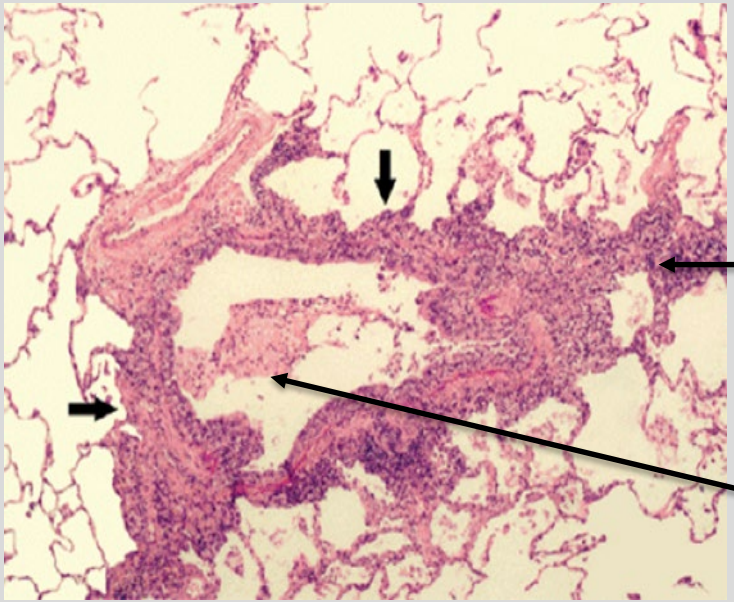
Morphology

Respiratory epithelial injury occurs after attachment of Mp to resp mucosa

Lesions may be patchy or may involve whole lobes bilaterally or unilaterally

Mp secretes peroxide: ulceration of mucosa.

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Histology

Inflammatory reaction, localized within walls of alveoli;

The alveolar septa are widened by an infiltrate: lymphocytes, histiocytes, and some plasma cells (short arrows).

Alveoli may show intra-alveolar proteinaceous material

Health-care (Nosocomial) Pneumonia

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: pulmonary infections acquired during a hospital stay:

-in patients with severe underlying disease, immunosuppression, prolonged antibiotic therapy, invasive access devices such as intravascular catheters.

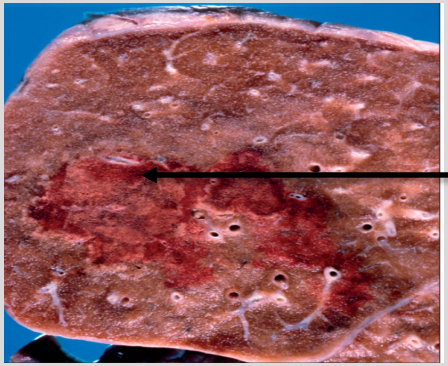
-patients on mechanical ventilation are at particularly high risk.

These patients have a higher mortality than those with community-acquired pneumonia.

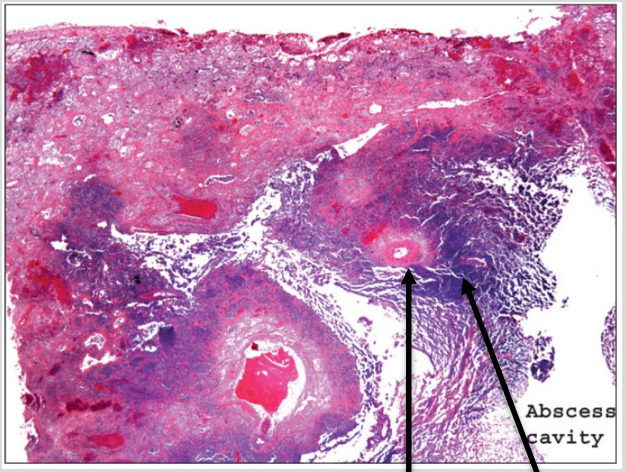
Methicillin-resistant S. aureus: Bronchopneumonia

Pseudomonas aeruginosa

In patients with cystic fibrosis, emphysema and long-term hospitalizations



Macro P. aeruginoas:
Erythematous patch surrounded by a tan rim.
Cavitation will develop and may result in abscess formation



Micro P aeruginosa: Blue areas: neutrophilic debris surrounding large vessels.

Aspiration Pneumonia

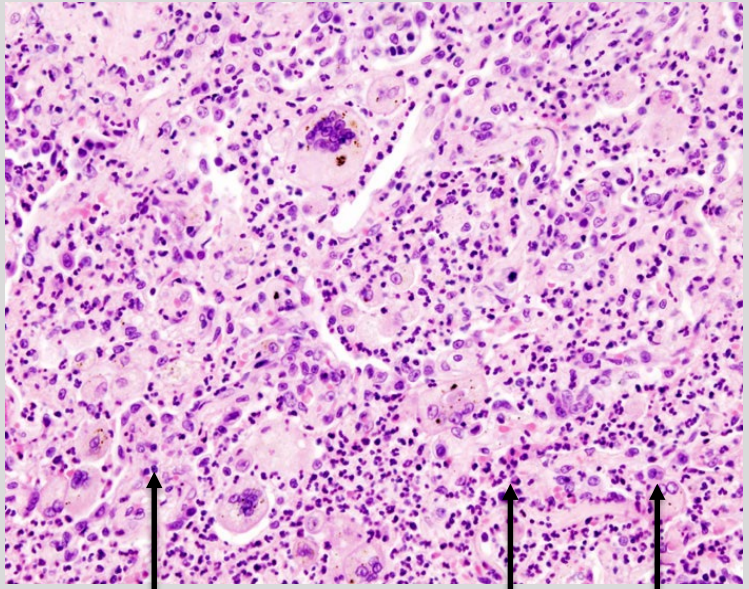
Aspiration pneumonia	Aspiration of oropharyngeal or gastric contents → pulmonary infection. Risk factors: altered mental status (↓ cough reflex or glottic closure), dysphagia, neurologic disorders (eg, stroke), invasive tubes (eg, nasogastric tube).	Presents days after aspiration event in dependent lung segment. More common in RLL if sitting up and RUL if lying down due to bronchial anatomy. Can progress to abscess. Aspiration (chemical) pneumonitis—presents hours after aspiration event. Due to gastric acid–mediated inflammation. Presents with infiltrates in lower lobe(s) and resolves with supportive treatment.

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It is partly chemical due to the irritating effects of gastric acid
It is partly bacterial from inoculation of the the oral flora

More than one organism is recovered on culture, aerobes being more common than anaerobes.

Morphology:
Necrotizing lung inflammation: frequent cause of death.
In patients who survive we may see **lung abscess development**.



Micro: Inflammatory infiltrate: PMNs, Eosinophils, Lymphocytes.

Lung Abscess

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Aspiration of oropharyngeal content/Bronchial obstruction

- **Aspiration of infective material**: the most frequent cause: Bacteroides, Fusobacterium, and Peptococcus
- **Antecedent primary lung infection**: S. aureus, K. pneumoniae, and pneumococcus.
- **Neoplasia**: Secondary infection is common in bronchopulmonary segments obstructed by a primary or secondary malignancy (postobstructive pneumonia) .
- **Septic embolism**: thrombophlebitis, infective bacterial endocarditis on the right side of the heart
- **Miscellaneous**: Traumatic penetrations of the lungs; direct extension of suppurative infections from the esophagus, spine, subphrenic space, or pleural cavity

Clinical features –Cough and copious amounts of foul-smelling purulent sputum, fever Chest pain, and weight loss are common. Clubbing of the fingers and toes may appear.

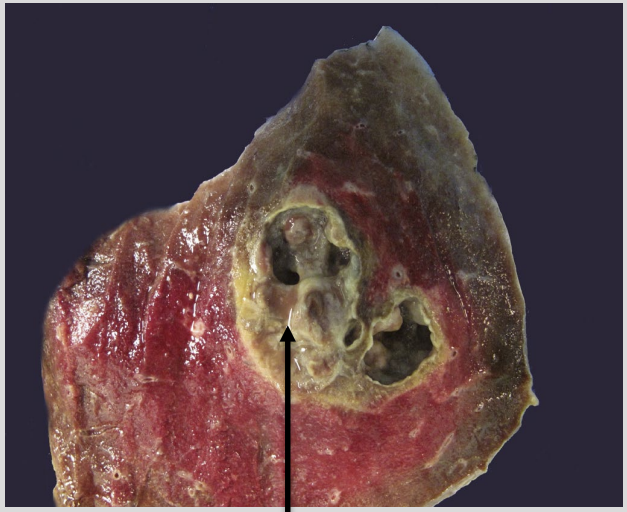
The diagnosis must be confirmed radiologically. **Abscess is in elderly: rule out an underlying carcinoma, which is present in 10% to 15% of cases.**

Therapy
-antibiotics

Complications : pleural effusion, hemorrhage, brain abscesses or meningitis from septic emboli, and (rarely) secondary amyloidosis.

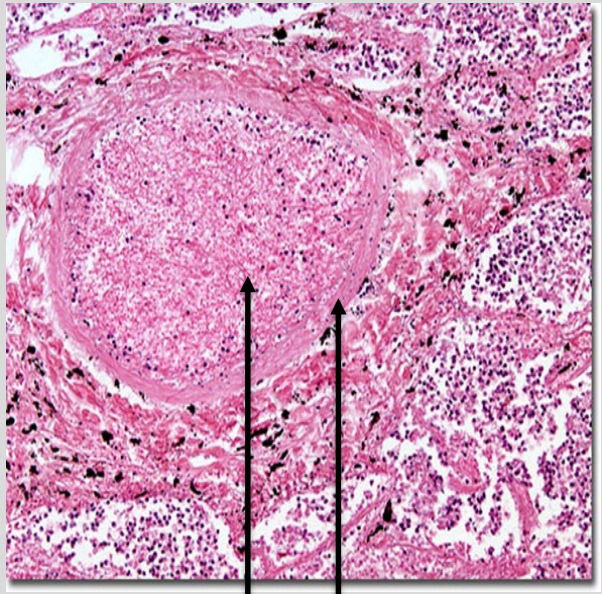
Lung Abscess Morphology

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Two abscesses

Aspiration: R side abscess as right main bronchus is vertical; single.
After pneumonia or bronchiectasis: abscesses are usually multiple, basal, and diffusely scattered



Histology: suppurative destruction of lung tissue within central area of cavitation; chronic abscesses have fibrosis.

Chronic Pneumonia

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It is most often a localized lesion in the immunocompetent patient, with or without regional lymph node involvement.

The inflammatory reaction is **granulomatous** and is caused by bacteria (Mycobacterium tuberculosis) or fungi.

Mycobacteria tuberculosis
-Granulomatous inflammation

Fungal:
Histoplasmosis
Blastomycosis
Coccidioidomycosis

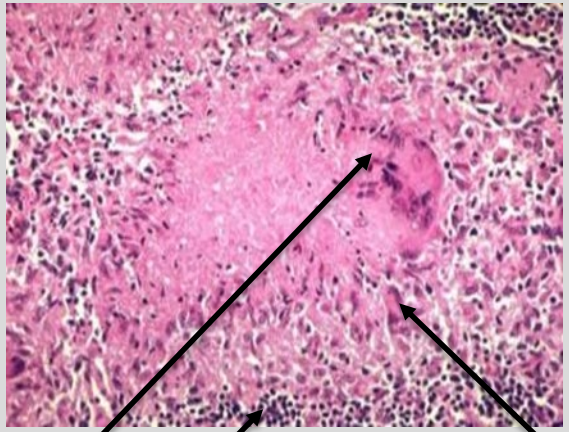
-Histo: **Ohio and Mississippi Rivers and Caribbean**
-Blasto: **Southeastern US**
-Coccidio: **Southwest and West US and Mexico**

Chronic Pneumonia: Mycobacterium tuberculosis

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Primary TB: Granuloma and Acute pneumonia which later becomes chronic

- It occurs in previously unexposed, unsensitized person: approx. 1 cm grey-white inflammatory consolidation in lung, **GHON FOCUS** (central necrosis)-**subpleural, often upper lobe nodule**;
- The micro-organisms drain to nearby lymph nodes which also caseate;
- The lung lesion + lymph node involvement = **GHON COMPLEX= GHON FOCUS plus lymphatic or hilar lymph node involvement**
- Most common location for Ghon complex: **lower lobe or anterior segment of the upper lobe.**

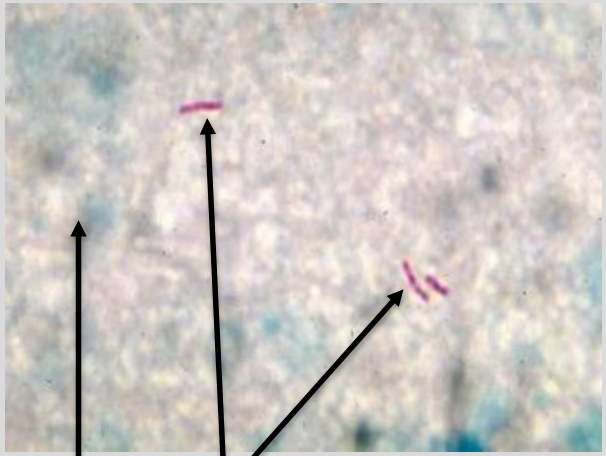


Histology: elongated/epithelioid histiocytes, giant cells with a peripheral cuff of lymphocytes and plasma cells.

Identification: Sputum smear stained with carbolfuchsin –Ziehl-Neelson method
Mycobacteria are acid-fast (retain stains when treated with acid and alcohol) because of waxy cell wall made of mycolic acid

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Gold standard for dx: culture



Acid fast bacili : Ziehl-Neels/Carbolfuchsin coloration
Degenerated cells

Key elements of pathogenesis:
-M. tuberculosis morphological changes can be explained as the outcome of a Th1 response.
-CD4 lymphocytes aka Helper T cells: Th1 and Th2 lymphocytes

Macrophages are primary cells infected and after 3 weeks, a **Th1 response** (produce IFN-gamma) against organisms occurs which activates macrophages to kill the mycobacteria;

However, at the same time a Th1 response also promotes **granuloma formation and necrosis:** hypersensitivity response leads to tissue destruction.

Secondary (Reactivation) Pulmonary tuberculosis

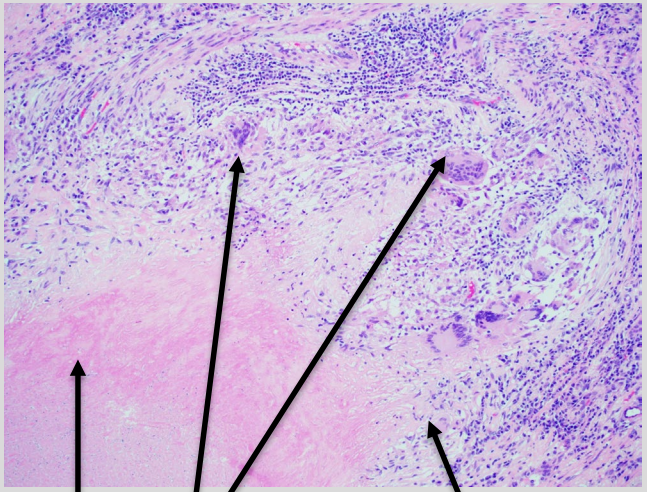
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Represents M. Tuberculosis in an immunocompetent individual.
It is also known as Post-primary Tuberculosis.
Only 5% to 10% of patients with primary tuberculosis will develop secondary form.

It begins at the apex of lung as a small focus (<3 cm) which evolves into a **pneumonia** characterized by the presence of a **fibrotic scar**, **granuloma** and an area of **central caseation.**

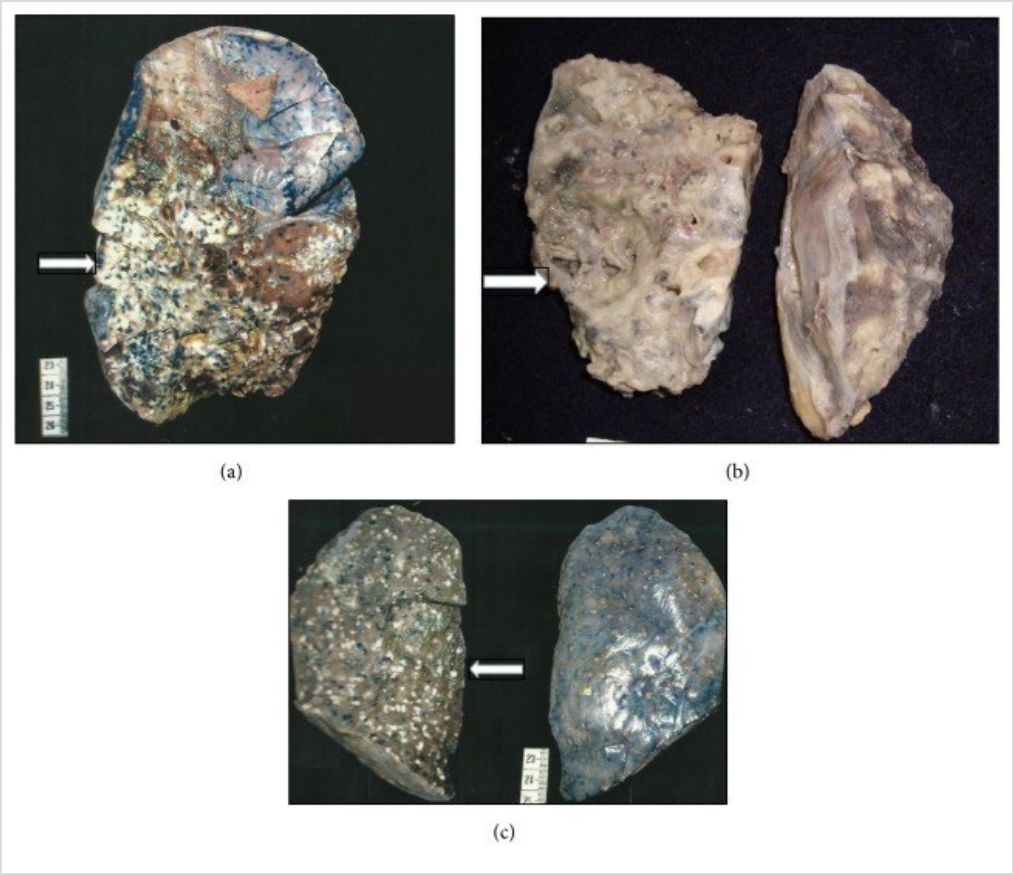
- **Not active-favorable evolution**: small area, no bronchial involvement
- **Active lesion**: acute pneumonia with cavitation
- Multiple foci coalesce and create confluent areas which communicate with a bronchus.

There is controversy surrounding this lesion as some authors suggest that these lesions represent exogenous reinfection.



Histology: granuloma-epitheliod histiocytes and giant cells around a center of caseous necrosis which contains acid fast organisms.

Morphology of secondary pulmonary tuberculosis



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Morphological patterns on gross examination of lung:

a. fibrocaseous: typical aspect

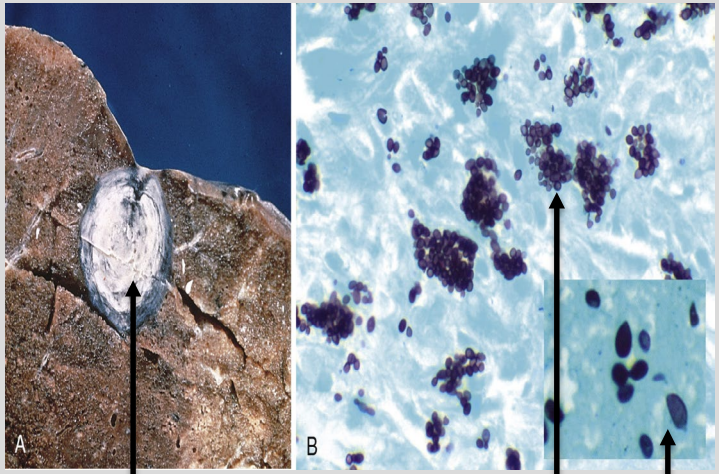
b. fibrocavitary tuberculosis with bronchiectasis (destruction and widening of the bronchi)

c. multiple millet sized tubercles in miliary tuberculosis.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4819112/

Histoplasmosis-Histoplasma capsulatum

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A. Histoplasma granuloma of the lung.

B. Histoplasma capsulatum yeast forms fill phagocytes in the lung of a patient with disseminated histoplasmosis; inset shows high-power detail of pear-shaped, budding yeasts (silver stain).

It is acquired by inhalation of dust particles from soil contaminated with bird or bat droppings that contain small spores (microconidia)

Histoplasmosis :in immunocompetent individuals but more severe in those with depressed cell mediated immunity.

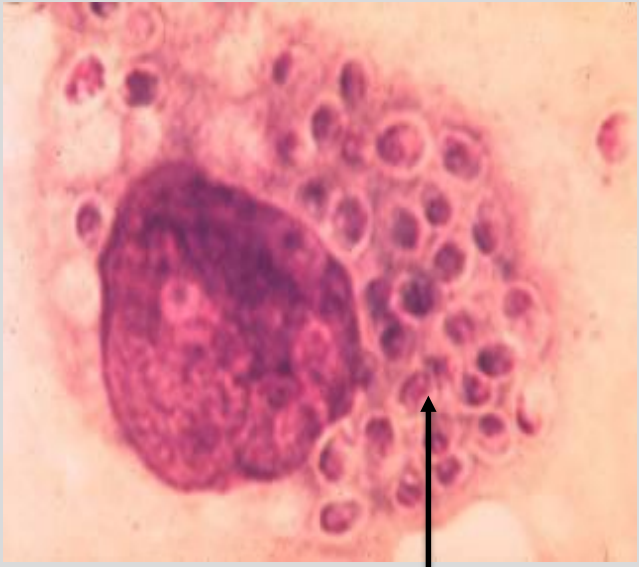
The pathogenesis of histoplasmosis is incompletely understood.
The portal of entry is virtually always the lung.

Macrophages ingest but cannot kill the organism without T-cell help

In individuals with adequate cell-mediated immunity, the infection is controlled by **Th1 helper T** cells that recognize fungal antigens and subsequently secrete IFN-γ, which activates macrophages and allow them to kill intracellular yeasts.

Histoplasmosis histology

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Macrophage containing numerous yeast cells of Histoplasma capsulatum. Dr. D. T. McClenan / CDC

Blastomycosis: Blastomyces dermatitidis

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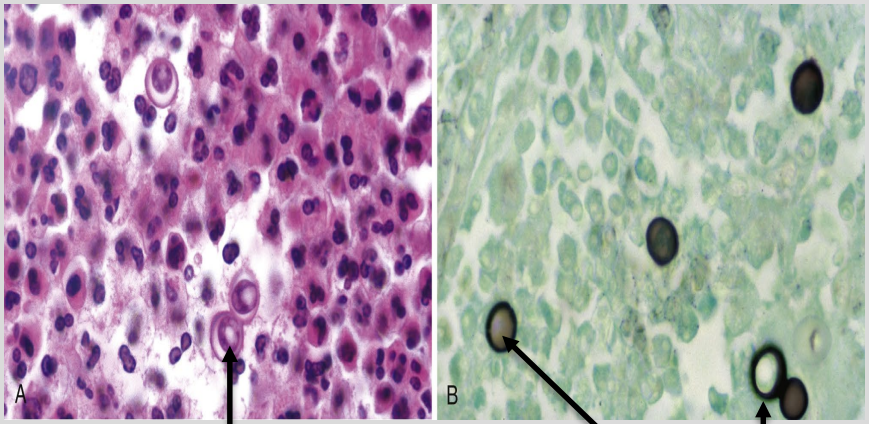
Pulmonary blastomycosis

The pneumonia most often resolves spontaneously, but it may persist or progress to a chronic lesion.

In the normal host, the lung lesions of blastomycosis are **suppurative granulomas.**

Macrophages have a limited ability to ingest and kill B. dermatitidis, and the persistence of the yeast cells leads to the recruitment of neutrophils and subsequent inflammation.

Disseminated blastomycosis
Primary cutaneous form: from direct inoculation of organisms into the skin.



A. Rounded budding yeasts, larger than neutrophils, are present. Note the characteristic thick wall and nuclei (not seen in other fungi). B. Silver stain.

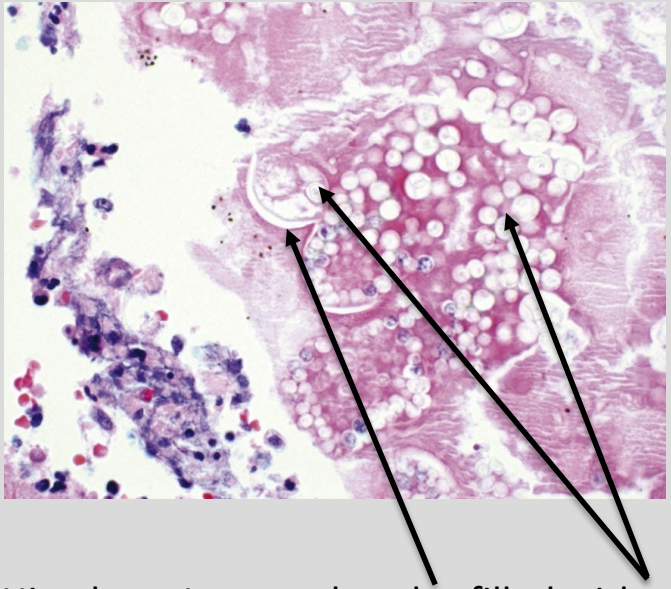
Coccidioidomycosis: Coccidioides immitis

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The patient develops a delayed-type hypersensitivity reaction

Approximately 10% of infected people develop lung lesions, and less than 1% of people develop **disseminated C. immitis infection**, which frequently involves the skin and meninges.

Infective C. immitis, when ingested by alveolar macrophages, block fusion of the phagosome and lysosome and so resist intracellular killing.



Histology: Intact spherules filled with small endospores

Pneumonia in the Immunocompromised Host

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Diffuse Infiltrates	Focal Infiltrates
Common	
Cytomegalovirus	Gram-negative bacterial infections
Pneumocystis jiroveci	*Staphylococcus aureus*
Drug reaction	*Aspergillus*
	Candida
	Malignancy
Uncommon	
Bacterial pneumonia	*Cryptococcus*
Aspergillus	*Mucor*
Cryptococcus	*Pneumocystis jiroveci*
Malignancy	*Legionella pneumophila*

The appearance of **a pulmonary infiltrate**, with or without signs of infection (fever), is one of the most common and serious complications in

- patients whose immune defenses are suppressed by disease
- immunosuppressive therapy for organ or hematopoietic stem cell transplants, chemotherapy for tumors, or irradiation.

In addition to the usual pathogens,
so-called opportunistic infectious agents, **many of which rarely cause infection in normal hosts**, can cause pneumonia

Mortality from these opportunistic infections is very high.

Pulmonary Infections in HIV patients

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Bacterial lower respiratory tract infections caused by the “usual” pathogens are among the most serious pulmonary disorders in HIV infection: S. pneumoniae, S. aureus, H. influenzae, and gram-negative rods.
-Bacterial pneumonias in HIV patients are often associated with bacteremia.

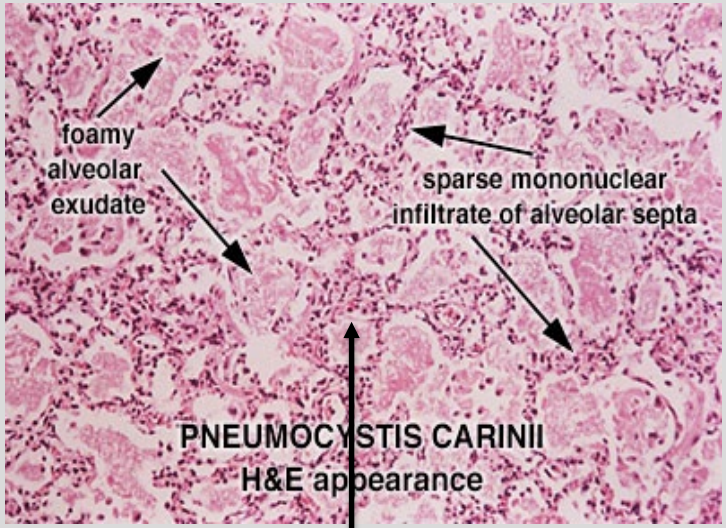
Bacterial/M. tuberculosis : at higher CD4+ counts more than 200 cells/mm 3
Pneumocystis: at CD4+ counts less than 200 cells/mm 3
CMV, fungal: at CD4+ counts less than 50 cells/mm 3

Not all pulmonary infiltrates in HIV-infected individuals are infectious in etiology:
Kaposi sarcoma, non-Hodgkin lymphoma, lung cancer.

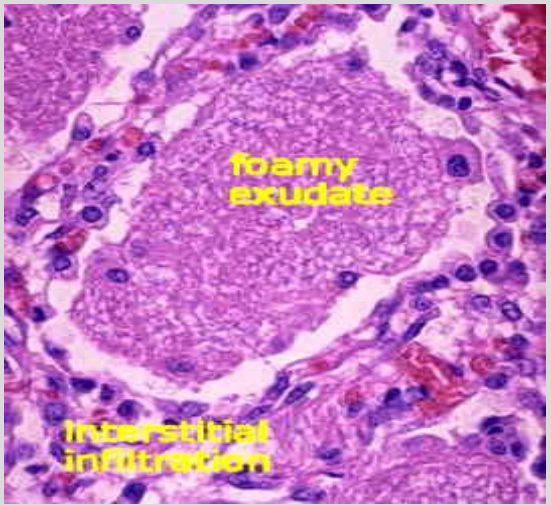
Pneumonia in Immunocompromised Host/ HIV Infection: Pneumocystis

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Tulane

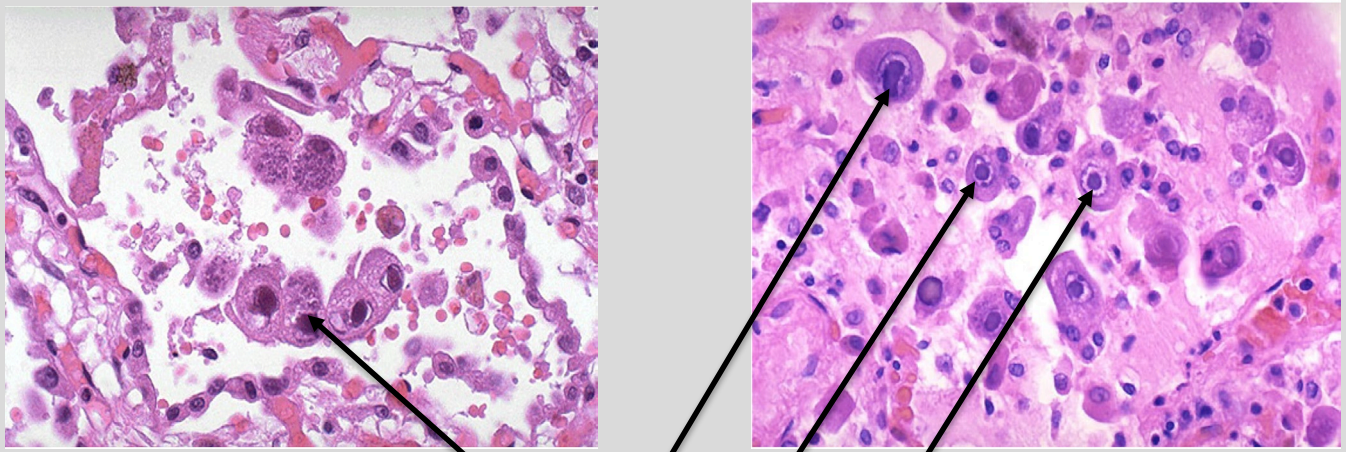


Pneumocystis carinii: extracellular pathogen; produces **an interstitial plasma cell pneumonia** with mild thickening of the aveolar septa, may progress to fibrosis. Alveoli fill with foamy vacuolated material.

Pneumonia in Immunocompromised Host/ HIV Infection: CMV

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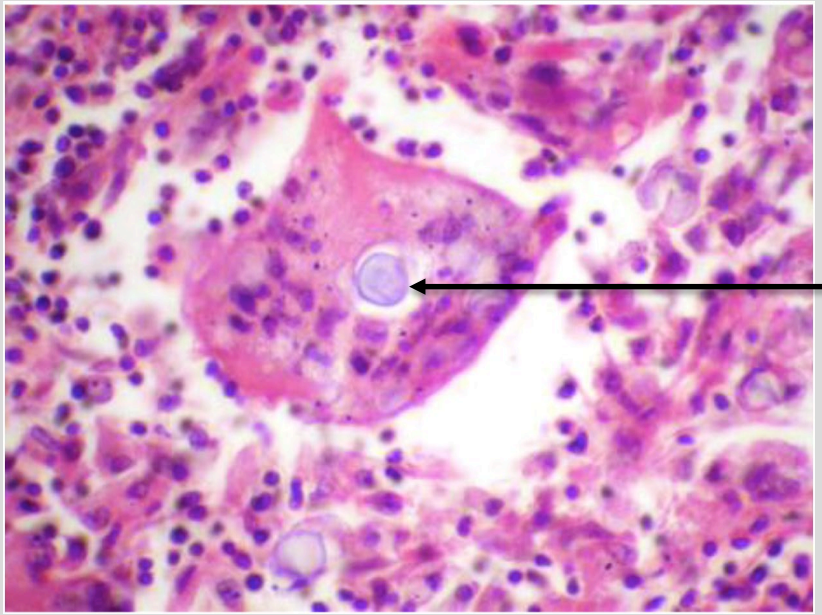


CMV: large cytomegalic cells with enlarged nuclei and purple intranuclear inclusion surrounded by a clear halo

Pneumonia in Immunocompromised Host/ HIV Infection: Cryptococcus

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Cryptococcus neoformans often produces a pneumonia; encapsulated yeast.



C. neoformans organisms have a clear capsule surrounding the pale blue nucleus

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